BSCS FINAL PROJECT

Software Requirements Specification - Phase 1

SAFORA (Smart Ride)

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Revision History

Version	Date	Reason for Change	Author
1.0	January 15, 2026	Initial Draft	Muhammad Zees, Ruhma Bilal

Abstract

SAFORA (Smart Ride) is an AI-powered ride-hailing platform implementing a proactive safety paradigm through: (1) Pink Pass - biometrically verified women-only mode with liveness detection, (2) Safety Sentinel - automated GPS monitoring detecting route deviations in real-time, and (3) Smart Driver Selection - AI-driven matching optimizing distance, rating, and fairness. The system uses MERN stack with Python/Flask AI microservice, employing Linear Regression for pricing, CNN for liveness verification, K-Means for demand heatmaps, and NLP for sentiment analysis.

1. Introduction and Background

1.1 Product (Problem Statement)

Current ride-hailing services in Pakistan suffer from three critical deficiencies:

Problem 1: Reactive Safety - Existing platforms rely on manual SOS buttons, ineffective when passengers cannot access their phones. No automated ride monitoring exists.

Problem 2: Gender Verification Gaps - Women-only modes use self-declaration without verification, allowing male drivers to use female accounts.

Problem 3: Inefficient Driver Allocation - Proximity-only matching causes monopolization by top-rated drivers and idle driver unemployment.

1.3 Scope

In Scope:

- User registration (Passenger/Driver/Admin) with CNIC verification
- Ride booking with real-time price estimation and GPS tracking
- Pink Pass enrollment with liveness detection (female passengers)
- Automated route deviation detection and dual alerts (admin + SMS)
- Smart driver matching using weighted algorithm
- Admin dashboard with heatmaps and sentiment analysis
- Payment (cash initially)

Out of Scope: Multi-city operations (Lahore only), third-party payment gateways, carpooling, advanced fraud detection.

1.4 Objectives

1. Implement automated route deviation detection with 95% accuracy
2. Achieve 98%+ liveness detection accuracy for Pink Pass
3. Reduce passenger wait time to <3 minutes
4. Provide AI price prediction with ±PKR 30 accuracy
5. Alert admins and emergency contacts within 30 seconds of anomaly

1.8 Completeness Criteria

- All user roles can register, authenticate, and perform core functions
- Pink Pass verification completes in <15 seconds
- Safety Sentinel monitors 100% of active rides
- Liveness detection: >95% TPR, <5% FPR
- Price prediction: $R^2 > 0.85$
- API response time <500ms for 95% of requests

- >80% code coverage for unit tests

1.10 Related Work

Feature	Uber	Careem	InDrive	SAFORA
Automated Route Monitoring	✕	✕	✕	✓
Biometric Gender Verification	✕	✕	✕	✓
Proactive Alerts	✕	✕	✕	✓
AI Fair Pricing	✓	Partial	✕	✓
Driver Fairness Algorithm	✕	✕	✕	✓

Academic Research: Zhang et al. (2021) showed ML models predict driver anomalies with 92% accuracy using GPS data. Patel et al. (2020) demonstrated CNN liveness detection achieves 98% accuracy in spoofing detection.

2. Overall Description

2.1 Product Features

Pink Pass: Two-layer verification (CNIC + liveness detection) using Haar Cascade for face detection and CNN for blink verification in <15 seconds.

Safety Sentinel: Monitors rides using point-to-line distance (Turf.js), triggering alerts if deviation >500m for >30 seconds or suspicious stops >5 minutes.

Smart Driver Matching: Weighted algorithm: Score = 0.50×Distance + 0.30×Rating + 0.20×Fairness where Fairness = IdleTime/TotalOnlineTime (last 4 hours).

Price Prediction: Linear Regression using 7 features (distance, duration, time, day, demand, area, traffic) with $R^2 > 0.85$.

Admin Dashboard: Real-time heatmaps (K-Means clustering), safety alerts, sentiment analysis (TextBlob), and analytics.

2.2 User Classes

Passenger: Individuals seeking rides. Basic smartphone proficiency. Goals: quick booking, safety, fair pricing. Usage: 3-10 rides/week.

Driver: Vehicle owners providing service. Moderate tech proficiency. Goals: consistent requests, maximize earnings. Usage: 4-8 hours online/day.

Administrator: Platform managers. High proficiency. Goals: real-time monitoring, dispute resolution, analytics. Usage: Full-time.

2.4 Design Constraints

Architecture: Microservice (Node.js/Express + Python/Flask AI service via REST APIs)

Mobile: React Native (iOS 13+, Android 8.0+)

Real-time: Socket.io (GPS updates every 5 seconds)

Data Privacy: AES-256 encryption for CNIC, 30-day location retention

Timeline: 6-8 months, 2 developers

Network: Must work on 3G/4G

Battery: <15% drain per hour for GPS tracking

3. Functional Requirements

3.1 Priority Use Cases

Critical (High Priority):

- UC-01: User Registration (Passenger)
- UC-02: User Registration (Driver)
- UC-03: User Authentication
- UC-04: Request Ride
- UC-05: Accept Ride (Driver)
- UC-08: Pink Pass Enrollment
- UC-09: Pink Pass Liveness Verification
- UC-10: Route Deviation Detection
- UC-11: Trigger Safety Alert

Important (Medium Priority):

- UC-12: Driver Matching Algorithm
- UC-13: Price Prediction
- UC-16: Admin Monitor Dashboard

3.2 Detailed Use Cases

UC-01: User Registration (Passenger)

Field	Description
ID	UC-01
Priority	High
Actors	Passenger, System
Pre-conditions	App downloaded
Post-conditions	Account created

Flow:

6. User enters: Name, Email, Phone, Password, CNIC
7. System validates format and performs OCR on CNIC photo
8. System sends OTP via SMS
9. User enters OTP
10. System creates account and displays success

Alternatives: Invalid format → error; Phone exists → error; OTP expired → resend option

UC-04: Request Ride

Field	Description
ID	UC-04
Priority	High
Actors	Passenger, System, AI Service, Driver
Pre-conditions	Logged in, GPS enabled
Post-conditions	Ride request sent to driver

Flow:

11. System detects GPS location and displays map
12. Passenger sets pickup and destination
13. System calculates route (Google Maps API)
14. AI Service predicts price using Linear Regression
15. System displays price breakdown
16. Passenger selects Standard/Pink Pass and confirms
17. System uses Smart Matching (UC-12) to find driver
18. System sends request to top-ranked driver

Alternatives: No GPS → error; Pink Pass not verified → redirect to UC-08; No drivers → retry message; Driver rejects → next driver (max 3 attempts)

UC-08: Pink Pass Enrollment

Field	Description
ID	UC-08
Priority	High
Actors	Female Passenger, System, AI Service
Pre-conditions	Registered as female
Post-conditions	Pink Pass activated

Flow:

19. Passenger navigates to Profile → Pink Pass
20. System displays consent form
21. Passenger accepts and camera activates
22. System displays "Look at camera and blink naturally"
23. Passenger positions face and blinks (5-second video)
24. System sends to AI Service (UC-09)
25. AI performs: Haar Cascade face detection + CNN blink detection
26. If successful: System updates pinkPass=true

Alternatives: Face not detected → retry with lighting; No blink → retry; Failed verification → "Try again or contact support"

UC-10: Route Deviation Detection

Field	Description
ID	UC-10
Priority	High
Actors	Safety Sentinel, System, Admin
Pre-conditions	Ride active
Post-conditions	Alert if deviation exceeds threshold

Flow:

27. Ride begins, system retrieves planned route (Google Maps polyline)
28. Safety Sentinel monitors every 5 seconds
29. System calculates: `turf.pointToLineDistance(currentLocation, route)`
30. If distance >500m: Start timer
31. If sustained >30 seconds: Trigger alert (UC-11)

Alternatives: Distance normalizes <30sec → reset timer; GPS lost → wait 60sec then flag

UC-11: Trigger Safety Alert

Field	Description
ID	UC-11
Priority	High
Actors	System, Admin, Emergency Contacts
Pre-conditions	Anomaly detected or SOS pressed
Post-conditions	Alerts delivered

Flow:

32. Anomaly detected or SOS pressed
33. System creates Alert record
34. System emits Socket.io event to admin dashboard
35. Admin receives pop-up notification
36. System retrieves emergency contacts
37. System sends SMS via Twilio: "SAFORA ALERT: [Name] may be in danger. Location: [Link]"
38. Admin can view live GPS feed

Alternatives: Twilio fails → retry 2x; No emergency contact → admin only

UC-12: Driver Matching Algorithm

Field	Description
ID	UC-12
Priority	High
Actors	System, AI Service
Pre-conditions	Ride request created
Post-conditions	Driver selected

Flow:

39. System queries drivers: status="online", within 5km
40. For each driver calculate:

   - Distance Score: 1 - (distance_km/5)

   - Rating Score: rating/5

   - Fairness Score: idleTime/totalOnlineTime (4hr window)

41. Compute: `0.50×Distance + 0.30×Rating + 0.20×Fairness`
42. Sort descending and send to top driver

Alternatives: No drivers 5km → expand to 10km; Pink Pass → filter gender="female" first

UC-13: Price Prediction

Field	Description
ID	UC-13
Priority	High
Actors	System, AI Service
Pre-conditions	Route calculated
Post-conditions	Price displayed

Flow:

43. System calculates distance (km) and duration (min) via Google Maps
44. System extracts: hour, day, demand level, area, traffic multiplier
45. POST to AI Service: `/api/ai/predict-price`
46. AI loads Linear Regression model (price_model.pkl)
47. AI predicts: `price = model.predict(features)`
48. Returns: `{price: 285, breakdown: {base:50, distance:170, time:44, surge:21}}`
49. System displays: "Estimated Fare: PKR 285"

Alternatives: AI unavailable → fallback Base + Distance×Rate; Unrealistic price → apply bounds (MIN=50, MAX=2000)

3.3 Requirements Analysis and Modeling

Required UML Diagrams (to be created):

50. **Use Case Diagram:** 3 actors (Passenger, Driver, Admin) with all 12+ use cases
51. **Entity-Relationship Diagram:** Entities - User, Driver, Ride, Alert, Review, EmergencyContact
52. **Class Diagram:** User (base), Passenger/Driver (extends), Ride, SafetySentinel, AIService
53. **Sequence Diagram:** Complete ride flow + Pink Pass verification flow

4. Non-Functional Requirements

4.1 Performance

- API response: <500ms (95% of requests)
- Price prediction: <2 seconds
- Driver matching: <3 seconds
- Liveness verification: <15 seconds
- Support 100 concurrent rides
- GPS updates: every 5 seconds
- Socket.io latency: <200ms

4.2 Safety

- 100% rides monitored by Safety Sentinel
- Deviation detection: <10 seconds
- Alert delivery: >99% success rate
- Dual alerts (SMS + dashboard) for redundancy
- Logs retained 90 days minimum

4.3 Security

- Password: min 8 chars (alphanumeric + special)
- JWT tokens: 24-hour expiration
- OTP: 6 digits, 5-minute validity
- HTTPS/TLS for all APIs
- AES-256 for CNIC storage
- Biometric data encrypted separately
- RBAC: Drivers cannot access Pink Pass data

5. Other Requirements

Localization: English/Urdu, PKR currency, Pakistan timezone (UTC+5)

Legal: Terms acceptance required, driver background checks, PDPB compliance

Usability: Material Design/HIG, max 3 taps for primary actions, color-coded alerts

6. Revised Project Plan

6.1 Timeline (22 Weeks)

Phase	Weeks	Key Deliverables
Setup	1-2	Repos, MongoDB, scaffolding
Backend Core	3-5	Auth, Ride APIs, Socket.io
AI Microservice	6-8	Pink Pass, Price model, Matching
Mobile Apps	9-12	Passenger/Driver apps, Maps
Safety Features	13-15	Safety Sentinel, Alerts, SMS
Admin Dashboard	16-17	Monitoring, Heatmaps, Analytics
Advanced AI	18-19	K-Means, Sentiment analysis
Testing & Deploy	20-22	UAT, Documentation, Presentation

6.2 Milestones

Milestone	Week	Deliverable
M1: Infrastructure Ready	2	All repos, DB connected
M2: Backend MVP	5	Ride booking functional
M3: AI Deployed	8	Pink Pass + Pricing working
M4: Mobile Beta	12	End-to-end ride flow
M5: Safety Live	15	Monitoring + Alerts tested
M6: Dashboard Complete	17	Full admin capabilities
M7: Production Ready	22	All features tested

7. References

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8. Appendices

Appendix A: Glossary

Term	Definition
Pink Pass	Biometrically verified women-only mode
Safety Sentinel	Background monitoring service for rides
Liveness Detection	AI verification of physical presence
Haar Cascade	Face detection algorithm
CNN	Convolutional Neural Network for image analysis
EAR	Eye Aspect Ratio for blink detection
K-Means	Clustering algorithm for heatmaps
Socket.io	Real-time bidirectional communication library
JWT	JSON Web Token for authentication
CNIC	Pakistan National Identity Card

Appendix B: IV&V Report

Project: SAFORA (Smart Ride)

Date: January 15, 2026

Reviewer: Sir Tanveer Ahmed

Requirement	Status
Functional requirements with use cases	✅Verified
Non-functional requirements	✅Verified
UML diagrams planned	⬜ Pending Phase 2
Timeline realistic	✅Verified
Technology stack justified	✅Verified
AI models defined	✅Verified

Advisor Signature: _____

Date: January 15, 2026

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